

CORE BANKING APPLICATION

Fixed Deposit (FD) — Deposits Page

TEST STRATEGY DOCUMENT

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Prepared By	Core Banking QA Manager
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1. Introduction

This Test Strategy Document defines the quality assurance approach for the Fixed Deposit (FD) module within the Core Banking Application. It establishes the testing framework, scope, approach, tools, and deliverables to ensure the FD Deposits page meets business, functional, integration, and regulatory requirements prior to production release.

This document is prepared for the System Integration Testing (SIT) phase and is governed by the QA Manager for Core Banking.

2. Objectives

- Validate end-to-end Fixed Deposit lifecycle — creation, renewal, premature withdrawal, and maturity processing
- Verify seamless integration between the FD module and the Internet Banking / Mobile Banking Portal
- Ensure accuracy of interest calculation, TDS deduction, and account statement generation
- Confirm regulatory and compliance requirements are met (RBI guidelines, KYC, AML)
- Validate functional, performance, and security quality risk areas
- Detect and report defects early to minimize cost of fix

3. Scope

3.1 In Scope

- Fixed Deposit (FD) account creation — new, joint, and minor accounts
- FD renewal — auto-renewal and manual renewal workflows
- Premature withdrawal — penalty calculation and processing
- FD maturity — principal + interest credit to linked savings/current account
- Interest calculation — simple and compound, quarterly/monthly/yearly payout
- TDS deduction and Form 15G/15H processing
- Internet Banking Portal integration — FD booking, viewing, and closure
- Mobile Banking Portal integration — FD management screens
- Notifications — SMS and email alerts for booking, renewal, and maturity
- Regulatory compliance — KYC linking, nominee details, AML checks
- Reports — FD statements, maturity reports, interest certificates

3.2 Out of Scope

- Savings Deposit, Recurring Deposit (RD), and Current Account Deposit modules
- Core Banking System (CBS) backend engine internals (black-box boundary)
- Third-party interest engine source code review
- Hardware infrastructure performance benchmarking
- UAT and Production deployment activities

4. Test Approach

4.1 Testing Methodology

A hybrid approach combining Manual Testing and Automation Testing will be adopted:

Testing Type	Approach	Tools / Frameworks
Functional Testing	Manual — exploratory + scripted	JIRA, Excel Test Cases
Regression Testing	Automated — full suite on every build	Selenium / Playwright + TestNG
API / Integration Testing	Automated — REST API contract testing	Postman / REST Assured
Performance Testing	Load & stress testing	JMeter / Gatling
Security Testing	Vulnerability & penetration testing	OWASP ZAP / Burp Suite
Accessibility Testing	WCAG 2.1 compliance checks	Axe / Wave

4.2 Testing Levels — SIT Focus

- Interface validation between FD module and Internet Banking / Mobile Banking Portal
- End-to-end transaction flow from portal UI through middleware to core banking and back
- Data integrity checks — values passed between systems are accurate and consistent
- Error handling — validate system behavior on downstream failures or timeouts
- Session and token management across integrated systems

5. Key Quality Risk Areas

5.1 Functional & Regression Risks

- Incorrect interest rate applied based on tenure and amount slab
- Auto-renewal triggering incorrectly or not triggering on maturity date

- Premature withdrawal penalty calculation errors
- TDS deduction applied at wrong threshold or not applied at all
- Linked account credit failure on FD maturity
- Regression breakage in existing FD features after new deployments

5.2 Performance & Load Risks

- FD booking page response time under high concurrent user load (>1000 concurrent users)
- Interest calculation batch job completing within the SLA window
- Mobile banking API response time for FD summary page < 2 seconds
- System throughput during month-end and quarter-end maturity processing peaks
- Database lock contention during bulk FD renewal operations

5.3 Security & Compliance Risks

- Unauthorized access to FD accounts via IDOR or broken object-level authorization
- Sensitive FD data (account number, balance) exposed in API responses or logs
- Session fixation or hijacking on Internet Banking portal
- TDS reporting data integrity — incorrect data in regulatory submissions
- KYC and AML validation bypass — customers with incomplete KYC booking FDs
- SQL injection and XSS vulnerabilities on FD creation forms
- Certificate pinning and HTTPS enforcement on mobile banking portal

6. Test Environment

- Environment: SIT — isolated from production, mirroring production configuration
- FD Module: Deployed build with latest approved changes
- Internet Banking Portal: SIT instance connected to FD SIT backend
- Mobile Banking Portal: Both Android and iOS builds pointed to SIT APIs
- Database: Masked/synthetic data representing all FD account types and tenures
- Test Data: Pre-seeded FDs at various lifecycle stages — new, matured, pre-closed, auto-renewed
- Network: Simulated latency environments for mobile testing using Charles Proxy

7. Test Deliverables

Deliverable	Description	Owner
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Test Strategy Document	This document	QA Manager
Test Cases (Manual)	Functional and negative test cases in JIRA/Excel	QA Lead
Automation Scripts	Regression suite — Selenium/Playwright	SDET
API Test Collection	Postman/REST Assured collection with assertions	SDET
Defect Reports	JIRA defect log with severity and priority	QA Team
Performance Test Report	JMeter results with response time graphs	Performance QA
Security Test Report	Vulnerability findings and CVSS scores	Security QA
SIT Sign-off Report	Final go/no-go recommendation with metrics	QA Manager

8. Entry & Exit Criteria

8.1 Entry Criteria (SIT Start)

- FD module code deployed to SIT environment and smoke tested
- Internet Banking and Mobile Banking portals connected and accessible in SIT
- Test data seeded and verified in SIT database
- Test cases reviewed and signed off by Business Analyst
- Test environment sign-off obtained from DevOps team

8.2 Exit Criteria (SIT Completion)

- 100% of planned test cases executed
- Zero open Critical or High severity defects
- 95% or above test case pass rate achieved
- All integration touchpoints with Internet Banking and Mobile Banking validated
- Performance benchmarks met — FD booking API < 3s, page load < 2s
- Security scan completed with no Critical/High CVEs unresolved
- Regression suite passing at 98% or above
- SIT sign-off approved by QA Manager and Business Stakeholder

9. Defect Management

Severity	Definition	Resolution SLA
Critical	System crash, data loss, FD booking failure, incorrect maturity payout	Fix within 1 business day
High	Wrong interest calculation, TDS error, portal integration failure	Fix within 2 business days
Medium	UI display issues, incorrect error messages, minor workflow gaps	Fix within 5 business days
Low	Cosmetic issues, label mismatches, minor UI inconsistencies	Fix in next sprint

Defect tracking tool: JIRA | Triage cadence: Daily stand-up during SIT cycle | Escalation path: QA Lead → QA Manager → Project Manager

10. Roles & Responsibilities

Role	Responsibilities
QA Manager	Test strategy, resource planning, SIT sign-off, stakeholder reporting
QA Lead	Test case design, defect triage, daily progress tracking
Manual QA Engineers	Functional and exploratory test execution, defect reporting
SDET / Automation Engineers	Automation script development, regression suite maintenance, API testing
Performance QA Engineer	Load test script creation, execution, and reporting
Security QA Engineer	Vulnerability scanning, penetration testing, compliance checks
Business Analyst	Requirements clarification, UAT coordination, test case review
DevOps / Infra Team	SIT environment provisioning, build deployment, access management

11. Test Metrics & Reporting

- Test Execution Progress: Daily — % test cases executed, passed, failed, blocked
- Defect Density: Total defects per test case executed
- Defect Leakage Rate: Defects found post-SIT / total defects found
- Automation Coverage: % of regression suite automated

- Performance Metrics: Response time P50/P90/P99, throughput (TPS), error rate
- Reporting Cadence: Daily status email + weekly QA dashboard to stakeholders

12. Assumptions & Constraints

12.1 Assumptions

- FD business rules and interest rate slabs are finalized and documented by Business team
- SIT environment will be stable and available during testing windows
- Internet Banking and Mobile Banking Portal builds will be available at SIT start
- Test data seeding scripts will be provided by the development team

12.2 Constraints

- SIT testing window is time-boxed as per the project release calendar
- Access to production data is not permitted — synthetic/masked data only
- Third-party payment gateway integration testing is limited to sandbox mode

13. Risks & Mitigation

Risk	Likelihood	Mitigation
Unstable SIT environment causing delays	Medium	Dedicated environment owner, daily health checks
Late delivery of Internet/Mobile banking build	High	Parallel API testing; stub/mock portal for FD backend testing
Incomplete or changing FD business requirements	Medium	Weekly BA alignment sessions; requirements freeze gate
Resource unavailability during peak SIT cycle	Low	Cross-training; backup QA resource identified
High defect count causing schedule overrun	Medium	Risk-based test prioritization; daily defect triage

— End of Test Strategy Document —

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